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Paleo- and Mesoproterozoic dyke swarms at Lake Ladoga area, NW Russia – paleomagnetic studies

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ABSTRACT: New paleomagnetic and rock magnetic data are presented from Paleoproterozoic 1.8 Ga shoshonitic intrusions and associated NW-SE trending lamprophyre dykes, and from undated NE-SW trending dolerite dykes at Lake Ladoga area in NW Russia. The shoshonitic intrusions and lamprophyre dykes are suggested to carry a primary 1.8 Ga remanent magnetization residing in titanomagnetite. In addition, they carry a partial remagnetization component that probably resides in titanomaghemite. The dolerite dykes show a single remanence component that is thought to be a primary thermoremanent magnetization. The paleomagnetic pole positions obtained from this component are Mesoproterozoic in age (ca. 1.54 Ga). This age corresponds to that of the Salmi rapakivi massif that is found near the study area. Thus, we interpret this magnetic component to be associated with widespread thermal and magmatic events associated with rapakivi magmatism and emplacement of associated dolerite dykes. Such events may also be responsible for the partial remagnetization of the shoshonites and lamprophyres.

1 INTRODUCTION

The Lake Ladoga area in NW Russia forms a key location of the Fennoscandian Shield in that it preserves a record of most of the significant geological events that the shield has experienced from the Archean to the Paleozoic. The main part of the studied area was formed during the Paleoproterozoic Svecofennian orogeny at ca. 1.9-1.8 Ga. Later on the Svecofennian rocks were intruded by Mesoproterozoic ca. 1.65-1.53 Ga rapakivi granites and associated dykes and volcanic rocks. In this study, we present paleomagnetic results from 1.8 Ga post-collisional shoshonitic intrusions (Eklund et al. 1998) and coeval lamprophyre dykes and from dolerite dykes that are thought to be related to the 1.54 Ga rapakivi granites. Geochemical, petrological and isotope age data obtained from the same sites as paleomagnetic data will be reported elsewhere (see Shebanov & Eklund 2005, Shebanov et al. 2006). The main aim of the paleomagnetic studies was to isolate the primary remanent magnetizations of the 1.8 Ga dykes and intrusions in order to obtain new reliable data for constructing the Fennoscandian plate movement, and to facilitate studies on the Paleoproterozoic supercontinents. The other aim was to study possible effects of the intrusion of the Mesoproterozoic rapakivi granites on the remanent magnetization of Paleoproterozoic rocks.

2 GEOLOGY AND SAMPLING

The area west of Lake Ladoga in northwestern Russia (Fig. 1) is mainly composed of Paleoproterozoic ca. 1880-1975 Ma tonalitic migmatites and granitoids (Konopelko & Eklund 2003, Konopelko et al. 2005). The northen part of the area is delinated by Archean (more than

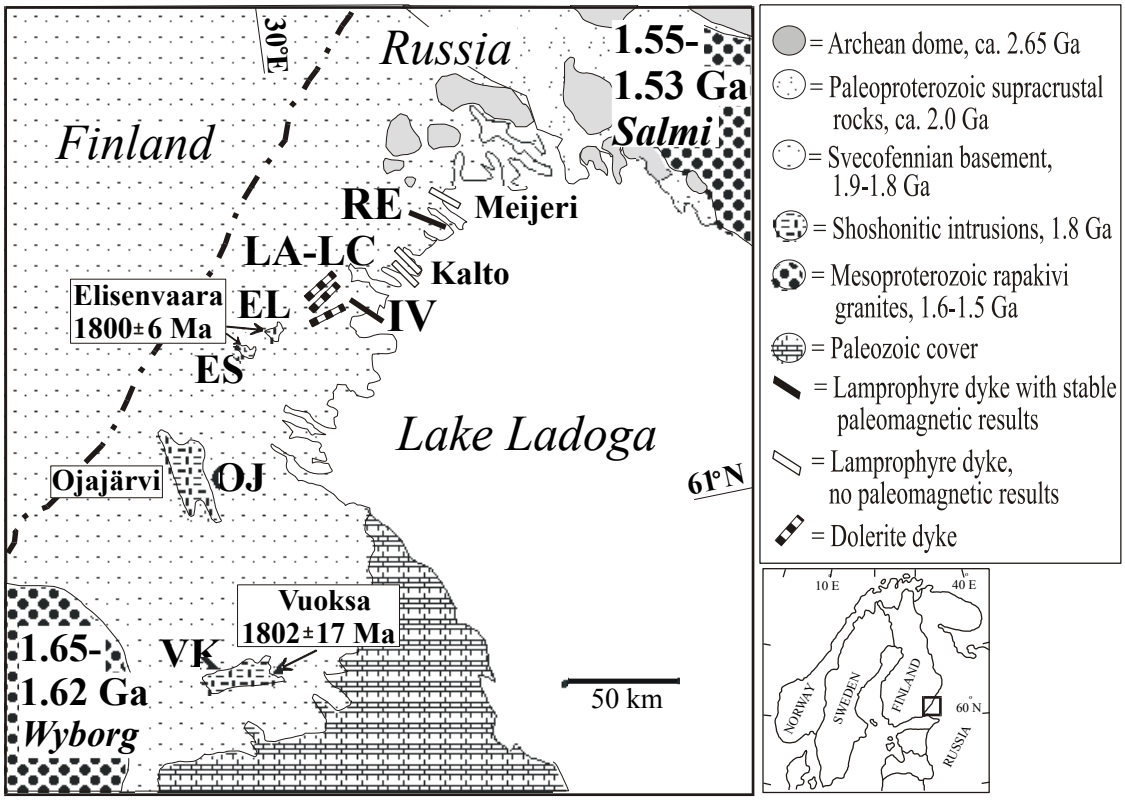


Figure 1. Location of the study area and sampling sites. Paleomagnetic results were obtained from the shoshonitic intrusions of Vuoksa (VK), Ojajärvi (OJ) and Elisenvaara (EL and ES), from the NW-SE trending lamprophyre dykes of Ivan's Island (IV) and Reuskula (RE) and from the NE-SW trending dolerite dykes LA, LB and LC. The lamprophyres of Kalto and Meijeri are also shown. The ages of the Salmi and Wyborg rapakivi batholits are indicated. Base maps by Koistinen et al. 2001.

2.65 Ga) and early Paleoproterozoic (ca. 2.0 Ga) rock types (Koistinen et al. 2001) and the 1547-1530 Ma (Amelin et al. 1997) Salmi rapakivi massif. In the southwestern part of the study area, the 1.65-1.62 Ga (Vaasjoki 1977) Wyborg rapakivi batholith is found.

In the post-collisional stage of the Svecofennian orogeny at ca. 1.8 Ga, the Svecofennian basement was intruded by shoshonitic magmas, ranging in composition from lamprophyres to granites. The shoshonitic intrusions represent magmas derived from enriched litospheric mantle, emplaced at upper crustal levels (Eklund et al. 1998, Konopelko & Eklund 2003). The intrusions form a conspicuous belt extending from Lake Ladoga area in NW Russia to Åland archipelago in SW Finland (Eklund et al. 1998). Paleomagnetic studies have now been carried out on three dated shoshonitic intrusions at Lake Ladoga area (Fig. 1); the Vuoksa (1802±17 Ma, U-Pb, zircon, Konopelko & Ivanikov 1996), Ojajärvi (1800-1805 Ma, K-Ar, amphibole, Ivanikov et al. 1996) and Elisenvaara (1800±6 Ma, U-Pb, zircon, Konopelko & Vaasjoki 2002, pers. comm.) intrusions. Paleomagnetic results from the intrusions are reported in more detail in Mertanen et al. (2005) and the main conclusions are briefly described in this paper.

The shoshonitic intrusions are genetically related to the lamprophyre dykes as they may be fractionated from lamprophyric melts (Eklund et al. 1998, Konopelko et al. 1998, Konopelko & Eklund, 2003). At least 7 lamprophyre dykes are presently known in the area (Vasilieva et al. 2002, Vasilieva & Frank-Kamenetsky 2003). Figure 1 shows schematically the locations of the dykes. U-Pb ages of the lamprophyres were determined on zircons with SIMS (Shebanov & Eklund 2005, Shebanov et al. 2006). Dating was carried out on three dykes in the Kalto Island, and on dykes of Ivan's Island (IV), Reuskula (RE) and Meijeri (Fig. 1). Isotopic dating of the lamprophyres was problematic because they commonly contained zircons that had been inherited from an Archean source. In the dyke of the Ivan's Island, no emplacement age was

obtained at all, as all the studied zircons are inherited and give only Archean ages. In the Reuskula lamprophyre the upper intersect gives an age of 2.75 Ga and the lower, an age of 1.8-1.7 Ga. The Meijeri dyke gives a population of inherited zircon with an age of 2.65 Ga, and an older Svecofennian age. The Kalto lamprophyres were most thoroughly analysed (4 dykes, 45 zircons) and they show different ages: 1) 1862±5 Ma, 2) 1896±7 Ma for a zircon core, and 3) 1777±33 Ma for the rim (Shebanov & Eklund 2005, Shebanov et al. 2006).

For paleomagnetic studies, samples were taken from the Vuoksa (VK) intrusion (9 samples), the Ojajärvi (OJ) intrusion (19 samples) and from the two pipe like bodies of the Elisenvaara (EL and ES) intrusion (11 samples). Seven lamprophyre dykes were sampled; three dykes in the Kalto Island, the Ivan's Island dyke (IV), the Reuskula (RE) and Meijeri dykes and a dyke that cross cuts the Elisenvaara intrusion. Unfortunately, the Kalto, Meijeri and Elisenvaara dykes did not carry stable remanent magnetizations. Consequently, paleomagnetic results were obtained only from two lamprophyres, the Ivan's Island dyke (18 samples) and the Reuskula dyke (5 samples). The trend of the lamprophyres is 112-115°. The 60 cm wide Reuskula dyke is vertical, but the 40-60 cm wide Ivan's Island dyke has a dip of 40° that is regarded to represent the original orientation acquired during the emplacement of the dyke (Vasilieva et al. 2002).

We also obtained paleomagnetic samples from three dykes (LA, LB and LC) from an undated dolerite dyke swarm (Fig. 1). On the basis of geochemical composition and the presence of anorthositic xenoliths and plagioclase phenocrysts, we consider these dykes to be genetically associated with the Mesoproterozoic rapakivi granites. The trend of the dykes is 75-60° and they are subvertical or vertical (dip 80-90°). The biggest of the dykes, the coarse grained dyke LB (12 samples) has a width of ca. 40 m, while the narrowest dyke LA (7 samples) has a width of 30-40 cm. Dykes LB and LC (10 samples) are mostly eroded away, but sampling could be carried out on chilled margins and on a restricted area in the center of dyke LB.

In order to carry out baked contact tests for the intrusions and the dykes, samples were taken also from the Svecofennian host gneisses. Unfortunately, the host rocks are very weakly magnetized and do not yield stable paleomagnetic results.

3 METHODS

Density and magnetic susceptibility were measured for all specimens before palaeomagnetic measurements. Remanence measurements were made using a 2G-Enterprises SQUID magnetometer and a Schoenstedt magnetometer. Demagnetizations were done both by alternating field (AF) up to peak field of 160 mT and thermally up to peak temperature of 680ºC. Mineralogical alterations were observed by monitoring the magnetic susceptibility after each thermal step. Remanence components were visually inspected by Zijderveld diagrams (Leino 1991, Zijderveld 1967) and the components were identified using principal component analysis (Kirshcvink 1980).

Magnetic carriers were identified by thermomagnetic analysis by using an Agico CS2-KLY2 Kappabridge, applying stepwise heating up to 700ºC. In order to further characterize the magnetic mineralogy, Lowrie-Fuller tests (Lowrie & Fuller 1971) were performed on selected specimens. The samples were first stepwise demagnetized in alternating fields up to 160 mT, and then given an anhysteretic remanent magnetization (ARM) in a peak AF of 200 mT and a biasing field of 0.1 mT. The ARM was then progressively AF demagnetized in peak fields up to 160 mT. The specimens were then pulse magnetized in a Molspin pulse magnetizer in fifteen steps between 25 and 1500 mT to produce SIRM. It was then progressively AF demagnetized in peak fields up to 100 or 160 mT.

4 ROCK MAGNETIC AND PALEOMAGNETIC RESULTS

4.1 *Rock magnetism*

Isothermal magnetization measurements for dykes and shoshonitic intrusions (Fig. 2) show a rapid increase up to about 200 mT and a total saturation well below 1 T, indicating magnetite or a low titanium titanomagnetite as the remanence carrier. Lowrie-Fuller tests (Lowrie & Fuller

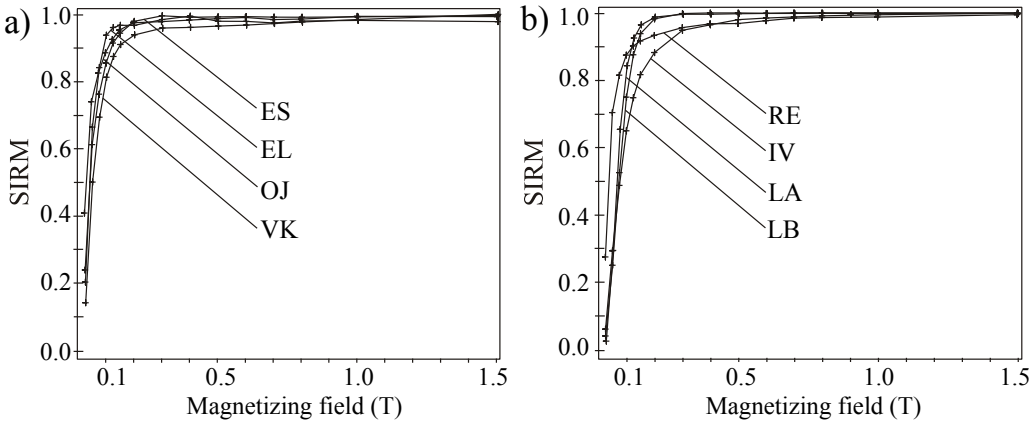


Figure 2. Saturation isothermal remanent magnetization (SIRM) plots for (a) shoshonitic intrusions and (b) lamprophyres (RE and IV) and dolerite dykes (LA and LB).

1971) show that in all rock types the remanence resides mainly in SD or PSD particles (Fig. 3). However, in samples from dyke IV, RE and OA, the ARM is weaker than IRM to AF demagnetization in high fields, suggesting that the remanence is carried by a mixture of finer SD/PSD and coarser MD grains.

Thermomagnetic analyses (magnetic susceptibility versus temperature, k-T) are shown in Figure 4. K-T curves for the Vuoksa and Ojajärvi shoshonitic intrusions (Fig. 4a,b) yield reversible heating and cooling curves with a Curie point of ca. 580°C, implying the presence of pure magnetite. However, the Elisenvaara intrusions (Fig. 4c,d) that are compositionally more heterogeneous than the Vuoksa and Ojajärvi intrusions, comprising a differentation series from ultramafic variations (Fig. 4c) to more acid rocks (Fig. 4d) show strongly irreversible heating and cooling curves, indicating mineralogical alterations during heating. Likewise, the lamprophyre and dolerite dykes (Fig. 4e-h) show conspicious increase of intensity in the cooling curves indicating considerable creation of magnetite during heating. In both specimens from the lamprophyre dykes IV and RE (Fig. 4e,f) there is a very slight decrease of intensity in the heating curves at the temperature of ca. 350ºC which may be an indication of the occurrence of titanomaghemite, titanomagnetite or pyrrhotite.

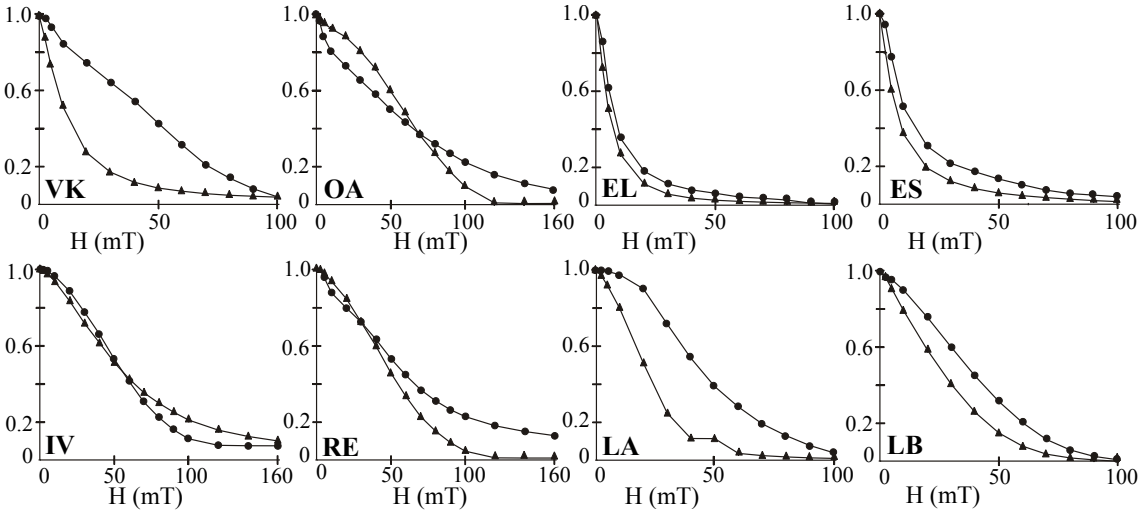


Figure 3. Lowrie-Fuller tests for samples from the shoshonitic intrusions VK, OA, EL and ES (upper row), of the lamprophyre dykes IV and RE, and of the dolerite dykes LA and LB (lower row). Circles denote AF demagnetization of ARM and triangles of SIRM.

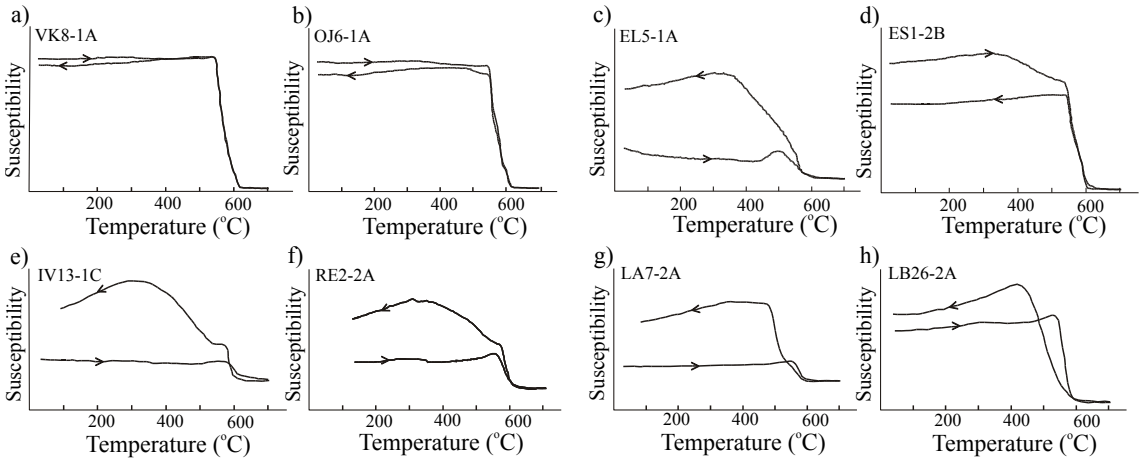


Figure 4. Thermomagnetic analyses from the shoshonitic intrusions (a-d), lamprophyre dykes (e,f), and dolerite dykes (g,h).

4.2 *Petrophysical properties*

All the studied itrusions and dykes that give stable paleomagnetic results are relatively strongly magnetized. The petrophysical properties; density, magnetic susceptibility, remanence inensity and Koenigsberger ratios (Q-values) are given in Table 1.

Table 1. Petrophysical properties of shoshonitic intrusions, lamprophyres and dolerite dykes at Lake Ladoga area.

	Density (kg/m^3)	Susceptibility ($x\ 10^{-6}$)	J of NRM (mA/m)	Q-value
Shoshonites				
VK	2 645 – 2 740	6 040 – 69 255	69 – 2 310	0.3 – 0.8
OJ	2 685 – 2 820	6 840 – 35 390	410 – 4 615	0.7 – 7.8
EL	2 875 – 3 030	620 – 46 985	10 – 1 270	0.2 – 0.6
ES	2 840 – 2 940	2 920 – 15 705	90 - 540	0.4 – 1.4
Lamprophyres				
IV	2 880 – 3 120	535 - 800	100 - 210	2.3 – 8.8
RE	2 810 – 2 850	510 – 60 410	20 – 890	0.3 – 5.1
Dolerites				
LA	2 875 – 2 890	7 805 – 14 900	2 560 – 3 510	5.9 – 8.3
LB	2 680 – 2 910	1 920 – 22 500	150 – 1 250	1.0 – 2.0
LC	2 720 – 2 885	650 – 18 350	1 – 1 180	0.04 – 4.8

Note. See abbreviations in Fig. 1. Susceptibility denotes magnetic susceptibility, J of NRM denotes intensity of natural remanent magnetization, Q-value is the Koenigsberger ratio.

4.3 *Paleomagnetism*

Paleomagnetic results from the dykes and shoshonitic intrusions are shown in Table 2. Examples of demagnetization behaviour of the shoshonitic intrusions are shown in Figure 5 and from the lamprophyre and dolerite dykes in Figures 6 and 7. The summary of the remanent magnetization components is shown in Figure 8.

4.3.1 Shoshonitic intrusions

The quartz monzonite of the Vuoksa intrusion (VK, Fig. 1) has a characteristic SE declination and intermediate negative inclination direction, component A_R (Table 2, Figs. 5 and 8a). The quartz monzonites of the Ojajärvi intrusion (OJ, Fig. 1) yields an opposite direction of NW declination and intermediate positive inclination, component A_N (Table 2, Figs. 5 and 8a). In

Table 2. Palaeomagnetic results from the shoshonitic intrusions and dyke rocks at Lake Ladoga area.

Comp./ Site	Lat, Long (°N), (°E)	B/N/n	Dec (°)	Inc (°)	α95 (°)	k	Plat (°N)	Plong (°E)	dp (°)	dm (°)	A95 (°)
Component A_R											
VK (Sh)	60.8, 29.7	1/*9/22	171.9	-46.3	8.1	41	56.4	223.2	6.7	10.4	9.7
Component A_N											
OJ (Sh)	61.2, 29.5	*3/17/43	345.6	43.5	8.4	216	53.0	231.1	6.5	10.5	8.1
IV (L)	61.4, 30.3	1/*12/27	342.7	32.3	3.7	139	44.6	233.7	2.4	4.2	3.3
RE (L)	61.6, 30.5	1/*4/8	349.5	40.2	8.6	116	50.9	225.8	6.2	10.3	6.0
Component S_R											
ES (Sh)	61.4, 29.8	1/*6/14	195.8	3.8	6.6	105	25.5	192.3	3.3	6.6	4.7
VK (Sh)		1/1/*2	195.7	11.8	-	-	22.1	192.8	-	-	-
OJ (Sh)		1/*2/3	210.5	27.2	-	-	10.6	179.4	-	-	-
IV (L)		1/*7/14	195.8	9.0	8.0	58	22.8	193.1	4.1	8.1	5.0
LA (D)	61.4, 30.1	1/*5/9	210.0	6.7	3.8	399	21.2	177.8	1.9	3.9	2.2
LB (D)	61.4, 30.2	1/*13/21	207.1	32.7	6.3	44	7.6	184.5	4.0	7.1	6.2
LC (D)	61.4, 30.2	1/*6/12	215.6	14.9	20.6	12	15.1	173.2	10.8	21.1	15.6
Mean (D)		*3/24/42	211.1	18.1	21.4	34	14.7	178.6	11.6	22.2	13.4
Component S_N											
RE (L)		1/*2/3	26.7	8.2	-	-	29.1	179.7	-	-	-

Note. Abbreviations for intrusions and dykes are given in Figure 1. (L), (Sh) and (D) denote lamprophyre, shoshonite and dolerite, respectively. Lat and Long give latitude and longitude of the sampling site. B/N/n are number of sites/samples/specimens used for mean calculations. * denotes the level used for mean calculation. $_N$ and $_R$ denote normal and reversed polarity, respectively. D and I are the mean declination and inclination. α95 is the radius of the circle of 95% confidence. k is the Fisher's (1953) precision parameter. Plat and Plong are the paleolatitude and paleolongitude for the Virtual Geomagnetic Poles. dp and dm are the semi-axes of the oval of 95% confidence. A95 is the radius of the circle of 95% confidence of the mean pole. Paleomagnetic results for the shoshonitic intrusions are shown in more detail in Mertanen et al. (2005).

both intrusions, a few samples show antipodal directions (Table 1, in Mertanen et al. 2005). The remanence resides in magnetite. The remanence directions of the Elisenvaara intrusion which is exposed as two pipe-like bodies, ES and EL (Fig. 1), differ from the directions of the Vuoksa and Ojajärvi intrusions. The other Elisenvaara body, ES, of a homogenous syenite, yields a characteristic remanence direction with a SW declination and low inclination, component S_R (Table 2, Figs. 5 and 8a). The other Elisenvaara body, EL (Fig.1), comprising of a variety of shoshonitic rock types, yields a characteristic remanence direction with a SE declination and steep negative inclination (component E, Table 1 in Mertanen et al., 2005). Because it is believed that the direction of this component reflects local tilting of the site, it is not used in any mean calculations (see Mertanen et al., 2003, 2005). In addition to component A, some samples of the Vuoksa (VK) and Ojajärvi (OJ) intrusions carry a smilar remanence component S_R (Table 2, Fig. 8b) as a partial overprint, as was observed in the Elisenvaara body ES (Table 2). However, the data are quite scattered and no mean value for component S_R was calculated for the shoshonites.

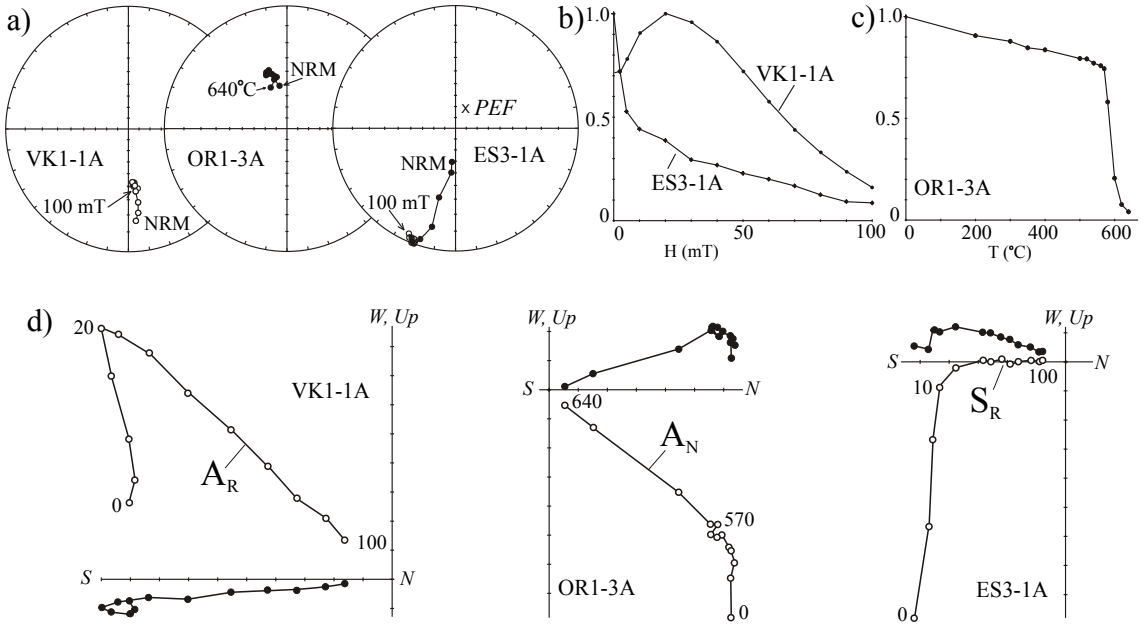


Figure 5. Examples of demagnetization behaviour of samples from the shoshonitic intrusions VK, OR and ES. Specimens VK1-1A and ES3-1A are demagnetized with AF and specimen OR1-3A with thermal demagnetization. a) stereoplots, b) intensity decay curves of AF demagnetization, c) an intensity decay curve of thermal demagnetization, d) orthogonal vector plots where open (closed) symbols denote vertical (horizontal) planes. Numbers refer to AF fields (mT) or temperatures (°C).

4.2.2 Lamprophyre dykes

Thermal demagnetization was more effective in isolating the characteristic components of the two magnetically stable lamprophyre dykes IV and RE (Fig. 1) than AF methods. By applying thermal demagnetization, two consistent remanence directions were isolated in all samples (Table 2 and Fig. 8b), while AF demagnetization mostly produced spurious directions which

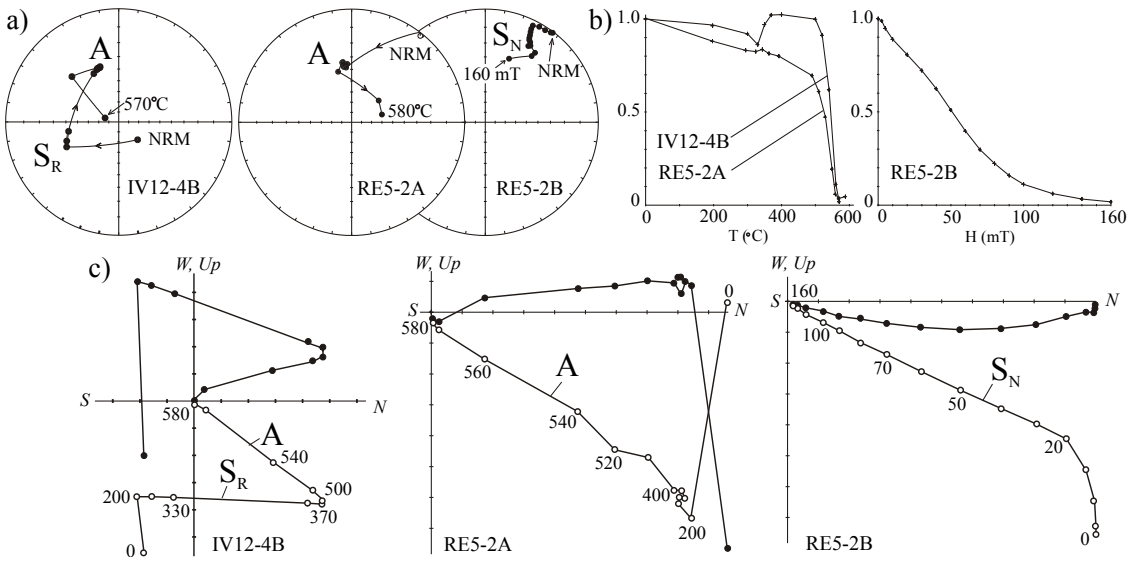


Figure 6. Examples of demagnetization behaviour of samples from the lamprophyre dykes IV and RE. Samples IV12-4B and RE5-2A are thermally demagnetized and sample RE5-2B is demagnetized with AF. For other explanations, see Fig. 5

deviate in different samples. The characteristic remanence direction A_N (Table 2, Figs. 6 and 8b) is isolated below the maximum unblocking temperature of ca. 580ºC suggesting that the remanence resides in magnetite. The remanence direction corresponds to component A_N obtained in the Ojajärvi intrusion. The lamprophyre dykes yield only single, normal polarity and no antipodal directions were observed. In addition to component A_N, both lamprophyre dykes carry remanence component S (Table 2, Figs. 6 and 8b) as a partial overprint. It is isolated in temperatures below 370ºC (see e.g. Fig. 6, specimen IV12-4B), suggesting titanomaghemite or titanomagnetite as the remanence carrier (see also Fig. 4). The direction corresponds to the partial overprint component S_R obtained in the Vuoksa and Ojajärvi intrusions, and as the characteristic component of the Elisenvaara intrusion ES. In the Ivan's Island dyke IV, the remanence is of reversed polarity (S_R) and in the Reuskula dyke RE, it is of normal polarity (S_N). Figure 6 shows an example of demagnetization behaviour from dyke RE where thermal demagnetization was more effective than AF method in isolating component A_N in one specimen (RE5-2A), and AF demagnetization was the best method to isolate component S_N in the other specimen (RE5-2B).

4.2.3 Dolerite dykes

The characteristic remanence component of the dolerite dykes show southwesterly declinations and shallow positive inclinations (Table 2, Figs. 7 and 8b) which are close to the direction of the S_R component of the lamprophyres and shoshonites. Like the lamprophyres, the dolerite dykes were more effectively demagnetized by thermal demagnetization than by AF methods, which in most cases showed odd behaviours. However, part of the AF demagnetizes samples showed extremely stable behaviour with a single component (see Fig. 7c). Dykes LA and LC have slightly lower inclinations than dyke LB. Dyke LB is 40 m wide and has cooled more slowly than the narrower dykes LA and LC, thus allowing some alteration to take place in the late stages of cooling. Thermal demagnetization of dyke LB indicate the presence of two magnetic phases that, based on unblocking temperatures, are probably low-titanium titanomagnetite (demagnetized between 520-570°C) and titanomaghemite (330-370°C). In the marginal parts of the dyke the remanence is unblocked within the whole temperature range of 330-570°C (see specimen LB21-4B, Fig. 7) thus evidencing that the maghemitization took place during cooling

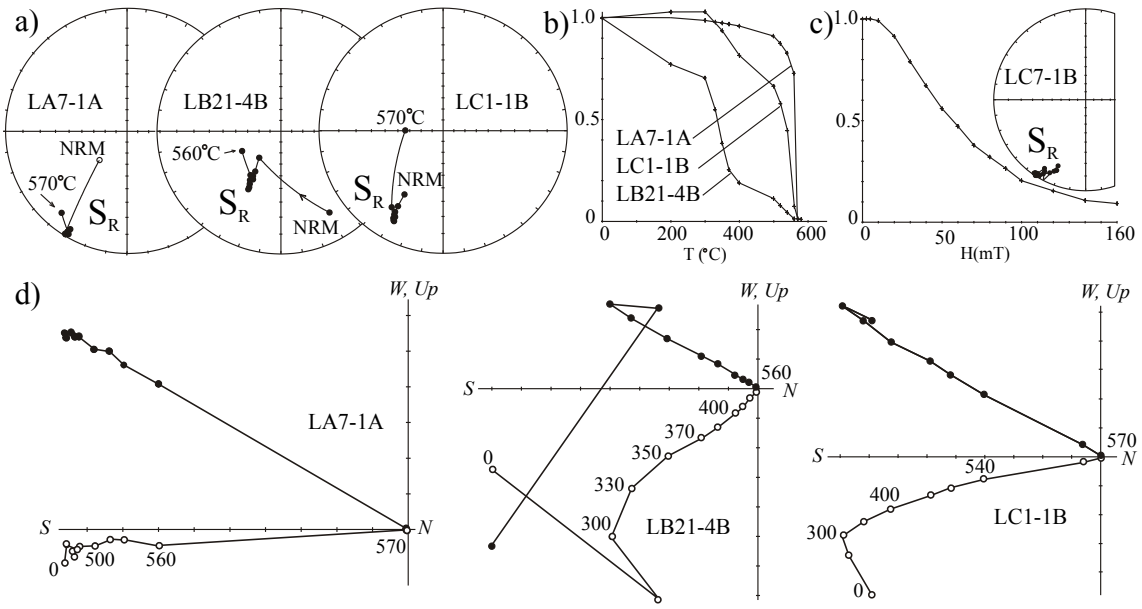


Figure 7. a-b, d) Examples of thermal demagnetization behaviour of samples from the dolerite dykes LA, LB and LC. For notes, see Fig. 5, c) Example of an AF demagnetization behaviour from dyke LC.

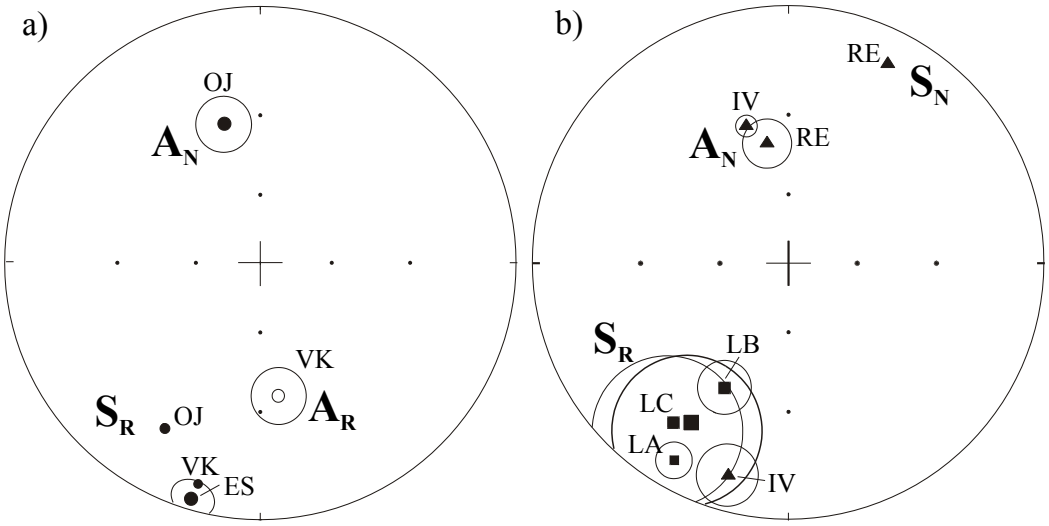


Figure 8. Site mean remanence directions of the shoshonites and dykes with the mean cone of 95% confidence. a) Circles, shoshonitic intrusions OJ, VK and ES, b) triangles, lamprophyre dykes IV and RE and squares, dolerite dykes LA, LB and LC. Closed symbols denote positive inclination and open symbol negative inclination.

of the dyke. In more central parts of dyke LB, about 1-2 m from the dyke margin, the titanomaghemite component has in general a slightly steeper inclination than the component residing in titanomagnetite.

5 DISCUSSION

In order to constrain the ages of magnetizations, paleomagnetic poles were calculated for the characteristic remanence components A and S (Table 2, Fig. 9). Although paleomagnetic field tests, like baked contact tests, were succesfull neither for the intrusions nor for the dykes due to weak remanence intensities and instability of remanence of the host gneisses, we believe that remanence component A is the primary remanence of the shoshonites and the lamprophyres. Since the intrusions were emplaced at upper crustal levels (Eklund et al. 1998, Konopelko & Eklund 2003) where the cooling of magma is relatively rapid, the U-Pb ages can be correlated with the timing of the remanence.

The shoshonitic intrusions are dated precisely (see section 2) and show ages of ca. 1.8 Ga. Also the SIMS datings of most of the lamprophyres give interpretable results (Shebanov & Eklund 2005, Shebanov et al. 2006). However, the age of some lamprophyres was poorly determined. The dating for the Ivan's Island dyke contained only inherited zircons with Archean ages, and no Svecofennian ages. However, based on the correspondence of the paleomagnetic poles from the shoshonites and the lamprophyres (Fig. 9a), both the Ivan's Island dyke and the Reuskula dyke are probably about the same age as the shoshonites. The paleomagnetic poles of component A are close to the known 1.88-1.83 Ga key poles of the Fennoscandian Shield. The 1.88 Ga key pole is obtained from gabbroic intrusions in central Finland and the 1.83 Ga key pole from the Haukivesi lamprophyre dykes in central Finland (see Buchan et al. 2000, Pesonen et al. 2003 for references). The result thus implies that during the Svecofennian orogeny, no significant plate movement took place. However, as the poles from the shoshonites show a little higher latitudes than the 1.88 and 1.83 Ga key poles, it is possible that at the post-collisional stage of the orogeny, the shield had moved to slightly higher paleolatitudes (Mertanen et al., 2003, 2005).

Poles from the secondary component S (Fig. 9a) plot among the 1.63-1.54 Ga key poles of the Fennoscandian Shield (see Buchan et al. 2000, Pesonen et al. 2003). The key poles are obtained from the diabase and quartz porphyry dykes associated with the rapakivi granites of the Åland and Onas rapakivi granites in southern Finland. Based on pole positions, which are

mainly based on data from a few samples, the exact age for pole S cannot be determined. However, the position of poles clearly indicates that magnetization S was acquired during Mesopropterozoic. Especially, the better defined poles from the Elisenvaara intrusion (ES) and from the Ivan's Island dyke (IV) are in agreement with the 1.54 Ga key pole of the Fennoscandian Shield (Fig. 9). The age of the Salmi rapakivi massif (Fig. 1) is 1.53-1.55 Ga, and, thus it represents a possible source for the remagnetization event. The Elisenvaara body was evidently totally remagnetized during the emplacement of the Salmi rapakivi massif (Fig. 1).

For the dolerite dykes (Fig. 9b) it is implied that remanence component S represents the primary magnetization of the dykes, although no field tests are available. The poles from the dolerite dykes compare rather well, although not fully matching with the 1.63 - 1.54 Ga key poles of the Fennoscandian Shield. Based on correspondence of pole S from the dolerite dykes to the key poles and on the basis of their compositional and petrological resemblance to the rapakivi granites, it is therefore likely that the dolerite dykes represent magmatism associated with the emplacement of the rapakivi granites. Emplacement of the dolerite dykes form a plausible additional source for the partial remagnetization of the 1.8 Ga old shoshonitic intrusions and lamprophyre dykes.

The paleomagnetic results from northewestern Russia document the vast effect of the rapakivi magmatism into the adjacent rocks. The Mesoproterozoic remagnetization was isolated as a partial overprint in rocks that are not in close proximity to the rapakivi granites. It is therefore implied that at Lake Ladoga area, the thermal effect of the rapakivi granites may have extended to the intermediate areas between the rapakivi batholiths. Previous paleomagnetic studies on shear and fault zones in southern Finland (Mertanen et al. 2004) have also demonstrated a reactivation and partial remagnetization of the crust during emplacement of the rapakivi granites. In this area the emplacement of the coeval dolerite dykes probably formed another source for reactivation.

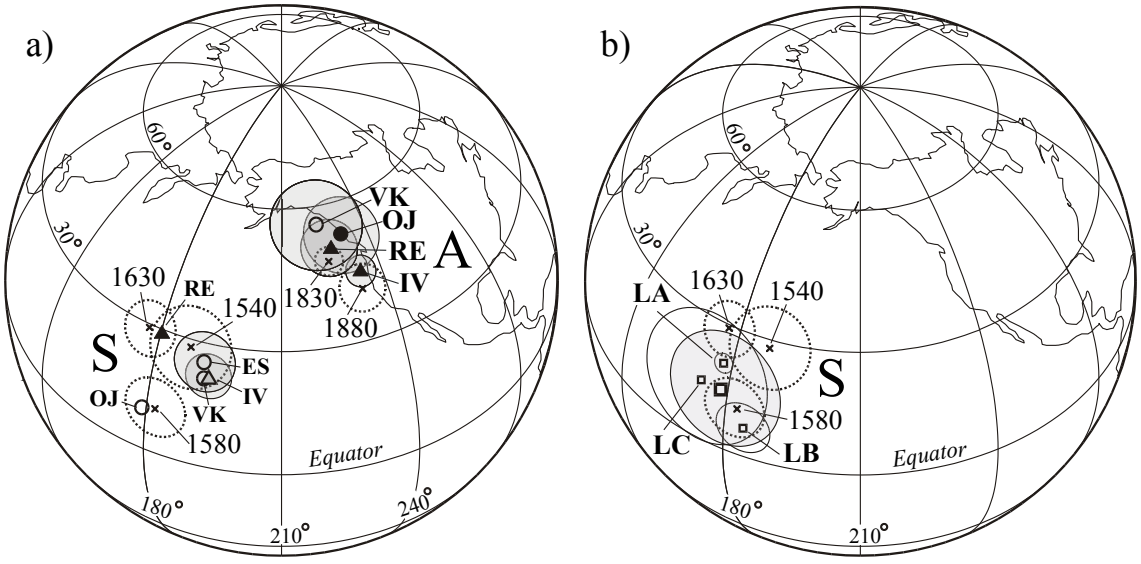


Figure 9. Paleomagnetic poles (VGP's) from a) the shoshonitic intrusions VK and OJ (circles), the lamprophyre dykes RE and IV (triangels) and from b) the dolerite dykes LA, LB and LC (squares). The Precambrian key poles of the Fennoscandian Shield are shown as crosses with dashed A95 confidence cones and ages (Ma). In a) the shaded A95 confidence cones are shown for the mean A poles VK, OJ, RE and IV that are regarded as primary, ca. 1800 Ma old, and for the mean S poles ES and IV that are regarded as secondary, ca. 1540 Ma old. The secondary poles RE, OJ and VK plot among the key poles of the ages of 1630-1540 Ga and were calculated for comparison, althoug the remanence component was obtained only in a few samples (therefore, no A95 confidence cones are shown). In b) the A95 confidence cone is shown for the mean pole from dykes LA, LB and LC, and is regarded as 1630-1540 Ma old.

6 CONCLUSIONS

Paleomagnetic results from well-dated 1.8 Ga shoshonitic intrusions are in accord with the results from the genetically related lamprophyre dykes. Both the shoshonitic intrusions and the lamprophyre dykes yield a stable remanence component (A) which is regarded to represent the primary remanent magnetization acquired ca. 1.8 Ga. In both rock types the A component is demagnetized below 580°C indicating magnetite as the remanence carrier. The paleomagnetic poles obtained from these rocks are close to the 1.88 and 1.83 Ga key poles of the Fennoscandian Shield, thus indicating that no extensive plate movement took place in the post-collisional stage of the Svecofennian orogeny.

The shoshonites and lamprophyres carry a secondary component (S) as an ancient overprint. It is demagnetized below 370°C thus suggesting that the remanence may reside in titanomaghemite. Comparison of the pole position to the key paleomagnetic poles, implies that the S remanence was acquired at ca. 1.54 Ga. The age of remanence is in agreement with the age of the nearby Salmi rapakivi granite massif. It is therefore concluded that the effect of the rapakivi magmatism has extended to the areas where no intrusions occur.

Another dyke swarm of undated dolerite dykes yield a corresponding remanence component S as what was obtained in the shoshonities and lamprophyres. In the dolerite dykes this remanence component is regarded to be of primary origin, acquired during cooling of the dykes at about the same time as the rapakivi intrusions. The emplacement of the dolerite dyke swarm thus forms another source for the partial remagnetization of the shoshonites and lamprophyres.

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